THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 9: **The Amazing World of Solutes, Solvents and Solutions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): A solution of sugar in water does not scatter a beam of light. Reason (R): The particles of the solute are of very small size. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. At 40 degrees Celsius, 100 g of water dissolves 64 g of potassium nitrate in a saturated solution. The mass percentage of the saturated solution is about: 1

(a) 64 per cent **(b)** 36 per cent **(c)** 39 per cent **(d)** 100 per cent

3. A liquid X mixes with water in all proportions while liquid Y does not. Which can be separated from water with a separating funnel? 1

(a) Y **(b)** X **(c)** both **(d)** neither

4. Assertion (A): A soft drink goes flat when left open. Reason (R): The solubility of a gas in a liquid decreases as pressure decreases. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. A solution is made by mixing 200 g of 10 per cent solution and 300 g of 20 per cent solution of the same solute. The percentage of the final solution is: 1

(a) 30 per cent **(b)** 16 per cent **(c)** 15 per cent **(d)** 14 per cent

6. Alloys such as brass are solutions in which: 1

(a) a gas in a liquid **(b)** a solid dissolves in a solid
(c) a gas in a gas **(d)** a liquid in a liquid

7. Assertion (A): Powdered sugar dissolves faster than sugar crystals. Reason (R): Powder has a smaller surface area. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. From 500 g of 20 per cent sugar solution, 100 g of water is evaporated. The new percentage is: 1

(a) 10 per cent **(b)** 25 per cent **(c)** 20 per cent **(d)** 30 per cent

9. A sample of tap water leaves white deposits on evaporation. This shows that it contains: 1

(a) pure water **(b)** oil

(c) dissolved solids **(d)** dissolved gases only

10. Assertion (A): The mass of a solution is equal to the sum of the masses of solute and solvent. 1
Reason (R): Mass is conserved during dissolving.

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. 100 g of a saturated solution at 60 degrees Celsius contains 37.5 g of solute. Find the mass percentage and the mass of water. 2

12. Explain why boiled and cooled water tastes flat. 2

13. Why does tea made with the same sugar but less water taste sweeter? 2

14. Calculate the mass of salt needed to make 750 g of a 6 per cent solution and the mass of water. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Compare the solubility of NaCl (36 g/100 g water at 20 degrees Celsius) and KNO3 (32 g/100 g water at 20 degrees Celsius). If 100 g water at 20 degrees Celsius is given 40 g of each, find the undissolved mass for each. 3

16. Explain how crystallisation purifies copper sulphate. 3

17. Why is sea water saltier than river water? 3

OR

A sample of 200 g of a 30 per cent sugar solution is diluted to 12 per cent. Find the water that must be added. 3

18. A colloid and a suspension are both cloudy. Give two tests to distinguish them. 3

OR

Explain why it is not correct to say that 'sugar disappears' when it dissolves. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. The solubility of a salt is 20 g per 100 g of water at 20 degrees Celsius and 50 g per 100 g of water at 70 degrees Celsius. 150 g of the saturated solution at 70 degrees Celsius is cooled to 20 degrees Celsius. Find the mass of solute at 70, mass of water, and mass of crystals formed. 5

OR

Explain how purified water is obtained from a salt solution by distillation, including the apparatus and principle, and why distilled water is used in labs. 5

20. Plan a project to find which of salt, sugar, baking soda and chalk powder is the most soluble in water at room temperature. Mention the procedure, controls and expected results. 5

OR

Describe the working of a water softener, a water purifier and the use of alum, based on dissolved and suspended impurities. 5

END OF QUESTION PAPER Total: 40 Marks